

A Hilbert-Range Subcase of the Finite-Dimensional Domain BPBp-RSE Question

FA/Banach discovery run `fa_banach_001`

2026-06-14

Source Question

The source paper is Helena del Río, *Bollobas-type theorems for range strongly exposing operators*, arXiv:2512.10442. It ends with the following question.

Suppose that X has finite dimension and Y is uniformly convex. Does the pair (X, Y) have the BPBp(-RSE)?

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Question 5.10. Is the BPBp-RSE equivalent to the BPBp-RSE for compact operators?

We finish with an open problem about the classical BPBp. A positive answer to this question would yield automatically the BPBp-RSE in virtue of Corollary 2.8.

Question 5.11. Suppose that X has finite dimension and Y is uniformly convex. Does the pair (X, Y) have the BPBp(-RSE)?

ACKNOWLEDGMENTS

The author wishes to express her gratitude to Sun Kwang Kim and Rubén Medina for suggesting the topic of this paper. She is also grateful to Mingu Jung and Óscar Roldán for providing valuable information and relevant references concerning specific parts of the work. This paper forms part of the author's PhD thesis,

This packet gives a positive answer for the Hilbert-range subcase. It is not a solution of the full uniformly convex range question.

Definitions Used

A pair (X, Y) has the Bishop-Phelps-Bollobas property for operators (BPBp) if, for every $\varepsilon > 0$, there is $\eta(\varepsilon) > 0$ such that whenever $T \in \mathcal{L}(X, Y)$, $\|T\| = 1$, and $x \in S_X$ satisfy

$$\|Tx\| > 1 - \eta(\varepsilon),$$

there are $S \in \mathcal{L}(X, Y)$ and $u \in S_X$ such that

$$\|S\| = \|Su\| = 1, \quad \|S - T\| < \varepsilon, \quad \|u - x\| < \varepsilon.$$

The source paper defines BPBp-RSE by requiring, in addition, that the approximating operator S be range strongly exposing. We use Corollary 2.8 of the source paper: if a subspace of $\mathcal{L}(X, Y)$ containing the rank-one operators has the classical BPBp, and $\text{str}(B_Y) = S_Y$, then it has the BPBp-RSE. Hilbert spaces are uniformly convex, hence every point of their unit sphere is strongly exposed.

Proof Intuition

The only point where the infinite-dimensional Hilbert range could cause a loss of compactness is the location of the finite-dimensional range $T(X)$. But Hilbert geometry removes this issue: every m -dimensional Hilbert subspace is isometric to the same model space ℓ_2^m . If $\dim X = n$, then an operator $X \rightarrow H$ only sees one of the finitely many Hilbert dimensions $1, \dots, n$. Taking the minimum of the known finite-dimensional BPBp moduli over those finitely many models gives a single modulus for the whole Hilbert space.

Theorem

Theorem 1. *Let X be a finite-dimensional real or complex Banach space and let H be a Hilbert space over the same scalar field. Then (X, H) has the BPBp-RSE.*

Proof. Let $n = \dim X$. We first prove the classical BPBp for (X, H) .

Fix $0 < \varepsilon < 1$. For each $1 \leq m \leq n$, let H_m denote an m -dimensional Hilbert space. Since both X and H_m are finite-dimensional, the finite-dimensional Bishop-Phelps-Bollobas theorem for operators of Acosta-Aron-Garcia-Maestre gives

$$\eta_m = \eta(X, H_m)(\varepsilon) > 0.$$

Put

$$\eta = \min_{1 \leq m \leq n} \eta_m > 0.$$

Let $T \in \mathcal{L}(X, H)$ satisfy $\|T\| = 1$, and let $x \in S_X$ satisfy $\|Tx\| > 1 - \eta$. Set $E = T(X)$. Then E is a Hilbert subspace of H of some dimension $1 \leq m \leq n$. Choose a surjective linear isometry $U : E \rightarrow H_m$. The operator $UT : X \rightarrow H_m$ has norm one and

$$\|UTx\| = \|Tx\| > 1 - \eta \geq 1 - \eta_m.$$

By the BPBp for (X, H_m) , there are $R \in \mathcal{L}(X, H_m)$ and $u \in S_X$ such that

$$\|R\| = \|Ru\| = 1, \quad \|R - UT\| < \varepsilon, \quad \|u - x\| < \varepsilon.$$

Define $S = U^{-1}R : X \rightarrow E \subset H$. Since U^{-1} is an isometry,

$$\|S\| = \|Su\| = 1, \quad \|S - T\| = \|R - UT\| < \varepsilon, \quad \|u - x\| < \varepsilon.$$

Thus (X, H) has the classical BPBp.

Finally, Hilbert spaces are uniformly convex. Hence $\text{str}(B_H) = S_H$. Applying Corollary 2.8 of the source paper to the full space $\mathcal{L}(X, H)$, which contains all rank-one operators, upgrades the classical BPBp to the BPBp-RSE. $\square$

Limitations and Novelty Check

This is only a Hilbert-range subcase of Question 5.11. The proof uses the fact that all finite-dimensional Hilbert subspaces of a fixed dimension are isometric. For a general uniformly convex range Y , the subspaces $T(X)$ of dimension at most $\dim X$ can have many non-isometric norms.

A tempting compactness argument over finite-dimensional range norms is not valid by itself: the BPBp modulus can degenerate along finite-dimensional Banach-Mazur convergent ranges. The

ACKLM example of two-dimensional polyhedral spaces Y_k with $\inf_k \eta(\ell_1^2, Y_k)(\varepsilon) = 0$ shows that Banach-Mazur compactness alone cannot give the desired uniform lower bound. A full proof for uniformly convex ranges would need a genuinely uniform-convexity-based stability argument, not merely finite-dimensional compactness.

Bounded literature checks on 2026-06-14 included exact and close searches for finite-dimensional domain BPBp with uniformly convex range, the source paper's final question, the ACKLM universal BPB-domain paper arXiv:1305.6420, and the finite-dimensional approach papers for ℓ_∞^3 and ℓ_∞^4 . These checks found known positive results for finite-dimensional ranges, ℓ_1^m domains, ℓ_∞^n domains with uniformly convex ranges, and $C(K)$ domains with uniformly convex ranges, but no full answer to Question 5.11. The Hilbert subcase may be regarded as an elementary consequence of the finite-dimensional BPBp theorem rather than a deep new theorem.

Human Review Recommendation

Check that the finite minimum of the finite-dimensional moduli is used only over Hilbert isometry classes H_m , $1 \leq m \leq \dim X$, and that the RSE upgrade is exactly Corollary 2.8 of the source paper. The packet should not be read as evidence that the arbitrary uniformly convex range case is settled.

References

- [1] M. D. Acosta, R. Aron, D. Garcia and M. Maestre, *The Bishop-Phelps-Bollobas theorem for operators*, J. Funct. Anal. 254 (2008), 2780–2799.
- [2] R. Aron, Y. S. Choi, S. K. Kim, H. J. Lee and M. Martin, *The Bishop-Phelps-Bollobas version of Lindenstrauss properties A and B*, Trans. Amer. Math. Soc. 367 (2015), 6085–6101; arXiv:1305.6420.
- [3] H. del Rio, *Bollobas-type theorems for range strongly exposing operators*, arXiv:2512.10442.